Chapter 11

Atomic Absorption Spectrophotometry

原子吸收光谱法

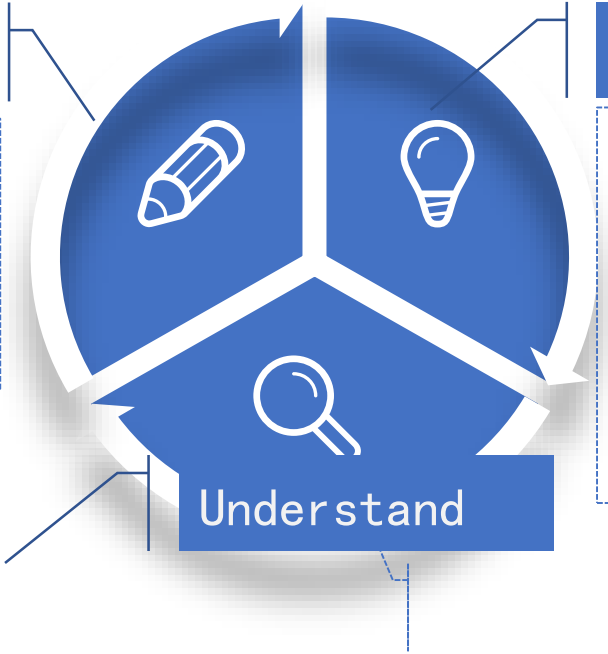
Contents

- Section 1 Overview
- Section 2 Basic Principles
- Section 3 Instrumentation
- Section 4 Quantitation

Learning Objectives

Master

- **Basic principles and quantitative methods of AAS**



Familiar

- **Components of the instrument; choice of experimental conditions; removal of interferences**

Understand

- **Features of AAS; causes of line broadening; instrument types**

Section 1 Overview

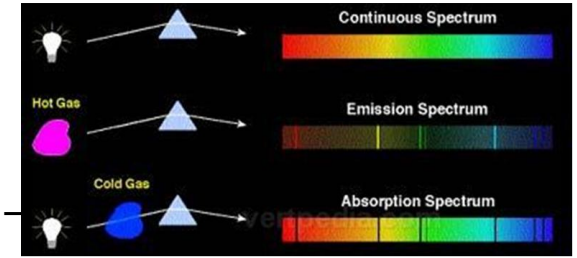
Section 1 Overview

方法名称	辐射能/激发方式	作用物质	检测信号
原子吸收光谱法	紫外、可见光	气态原子外层电子	吸收后的紫外、可见光
紫外-可见光谱法	紫外、可见光	分子外层电子	吸收后的紫外、可见光
红外吸收光谱法	红外光	分子振动	吸收后的红外光
核磁共振波谱法	0.1~900 MHz射频	原子核磁量子等	吸收
原子发射光谱法	火焰、电弧等	气态原子外层电子	紫外、可见光
分子荧光光谱法	紫外、可见光	分子	荧光（紫外、可见光）
磷光光谱法	紫外、可见光	分子	磷光（紫外、可见光）
化学发光法	化学能	分子	可见光
拉曼光谱法	红外光	分子振动	散射后的红外光

Section 1 Overview

Atomic Spectrometry and Molecular Spectrometry

- 原子光谱法Atomic Spectrometry
  - 气态原子或离子外层或内层电子能级跃迁所产生Produced by the energy level transitions of outer or inner electrons of gaseous atoms or ions
  - 明锐的彼此分离的谱线组成的线状光谱 Line spectra composed of sharp, well-separated spectral lines
  - Independent of the molecular state of the source
  - 与来源分子状态无关，确定元素组成和含量used to determine elemental composition and content



分子光谱（带状）

原子光谱（线状）

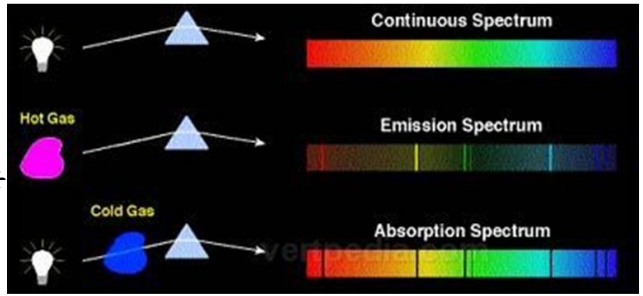
原子光谱（线状）

原子光谱和分子光谱

Section 1 Overview

Atomic Spectrometry and Molecular Spectrometry

- 分子光谱法Molecular Spectrometry
  - 分子与电磁辐射作用，产生能级之间的跃迁Interactions between molecules and electromagnetic radiation induce transitions between energy levels
  - 带状光谱，电子能级、振动能级、转动能级Band spectra, involving electronic, vibrational and rotational energy levels
  - 给出物质分子结构信息Provides information on the molecular structure of substances



分子光谱（带状）

原子光谱（线状）

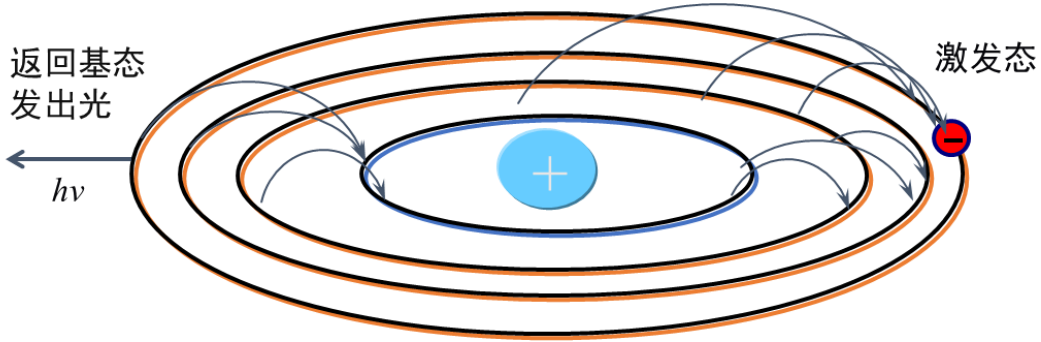
原子光谱（线状）

原子光谱和分子光谱

Section 1 Overview

原子吸收光谱法Atomic absorption spectroscopy, AAS

- It is a quantitative analysis method based on the absorption of characteristic electromagnetic radiation by the ground-state atoms of the analyte in the vapor.是基于蒸气中待测元素的基态原子对特征电磁辐射的吸收作用来进行定量分析的方法
- Atomic absorption spectra fall within the ultraviolet and visible regions of the spectrum原子吸收光谱位于光谱的紫外区和可见区

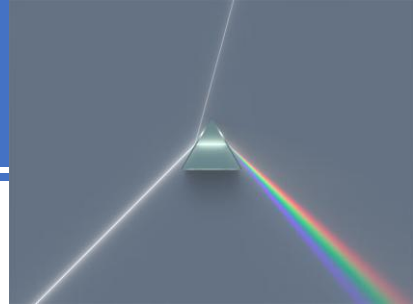


Section 1 Overview

原子吸收光谱法Atomic absorption spectroscopy, AAS

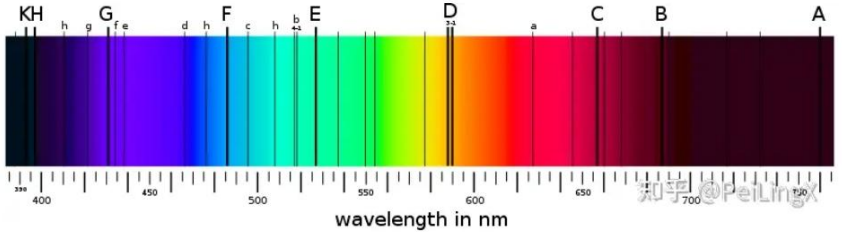
- Atoms are usually in the ground state. When the energy supplied externally exactly equals the energy difference required for the outer electrons of the atomic nucleus to transition from the ground state to an excited state, the extranuclear electrons will absorb light radiation of characteristic energy, transition from the ground state to the corresponding excited state, and thus produce atomic absorption spectra.通常情况下原子处于基态，当外界提供的能量恰好等于原子核外层电子由基态跃迁至激发态的能量差时，核外电子将吸收特征能量的光辐射，由基态跃迁到相应激发态，而产生原子吸收光谱

Section 1 Overview



发展历史Development History

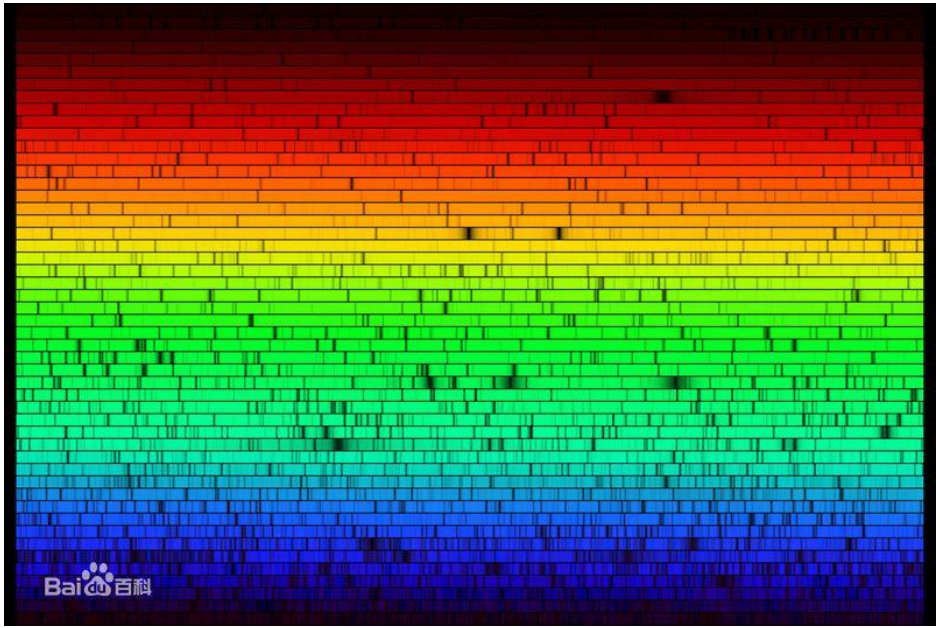
- In the 1660s — Isaac Newton: sunlight passed through a small aperture, and using a prism, he discovered that light is composed of monochromatic lights of different colors — the earliest form of spectral analysis 17世纪60年代——牛顿：太阳光通过小孔，用三棱镜发现光是由不同颜色的单色光组合而成——最原始的光谱分析
- In 1802, the British chemist W. H. Wollaston used a narrow slit and higher-quality prisms to observe the solar spectrum and discovered several dark lines.1802年英国化学家W.H.Wollaston，用狭缝+质量更好的棱镜，观察太阳光谱，发现几条暗线



Section 1 Overview

发展历史Development History

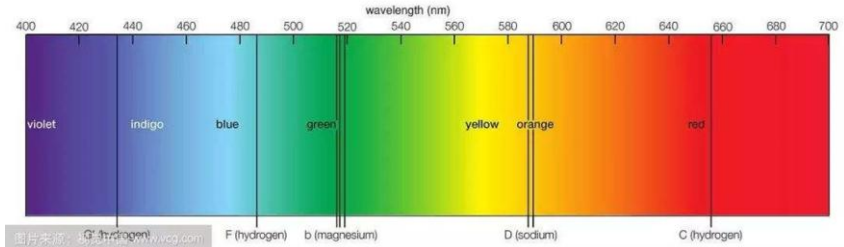
- In 1817, when J. Fraunhofer observed the solar spectrum, he discovered 576 dark lines in the spectrum — the Fraunhofer lines.
- 1817年J.Frauhofer观察太阳光谱时，发现光谱中有576条暗线——Frauhofer线（夫琅禾费线）



Section 1 Overview

发展历史Development History

- In 1859, while studying the flame spectra of alkali metals and alkaline earth metals, the German physicist G. Kirchhoff and chemist R. W. Bunsen discovered that when light emitted by sodium vapor passed through sodium vapor at a lower temperature, absorption of sodium light would occur. They concluded that the dark lines in the continuous solar spectrum resulted from the absorption of sodium radiation in the solar spectrum by sodium atoms in the Sun's outer atmosphere.1859年德国物理学家G.Kirchhoff和化学家R.W.Bunsen研究碱金属和碱土金属的火焰光谱时，发现钠蒸气发出的光通过温度较低的钠蒸气时，会引起钠光的吸收，断定太阳连续光谱中的暗线，正是由太阳外围大气圈的钠原子对太阳光谱中的钠辐射吸收的结果



Section 1 Overview

The main advantages of atomic absorption spectrophotometry
原子吸收分光光度法的主要优点

- ① high sensitivity; ② good selectivity; ③ high precision; ④ wide measurement range①灵敏度高；②选择性好；③精密度高；④测量范围广

The main limitations of atomic absorption spectrophotometry
原子吸收分光光度法的主要局限性

- ① It is difficult to realize the simultaneous determination of multiple elements, and a corresponding hollow cathode lamp is required for each element; ② The linear range of the working curve is narrow; ③ The detection capability for some elements is poor①难实现多元素同时测定，每种元素需要对应阴极灯；②工作曲线线性范围窄；③某些元素检出能力差

Section 2

Basic Principles of AAS

Section 2 Basic Principles

1 Quantum energy levels & diagrams量子能级与能级图

1. Motion of a single extranuclear electron核外一个电子的运动状态

With one outer electron, the state is described by four quantum numbers四个量子数

Principal quantum number主量子数n: 1, 2, 3, 4... (shell电子层)

Angular quantum number角量子数l: 0, 1, 2, 3... (n-1) (subshell电子亚层: s, p, d, f)

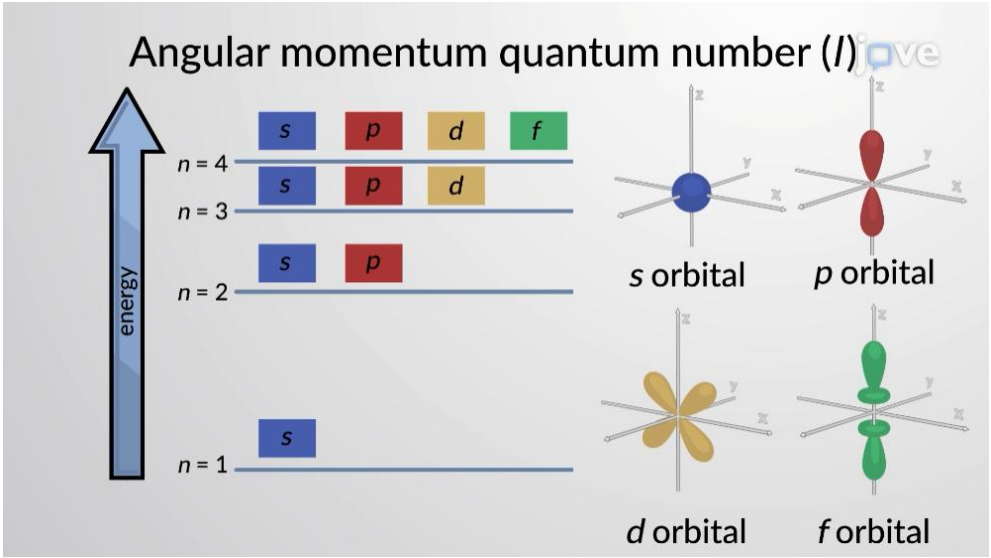
Magnetic quantum number磁量子数m: 0, ±1, ±2... ±l (orientation of the electron cloud电子云的伸展方向, e.g. px, py, pz)

Spin quantum number自旋量子数m_s: +1/2, -1/2

1 Quantum energy levels & diagrams量子能级与能级图

1. Motion of a single extranuclear electron核外一个电子的运动状态

With one outer electron, the state is described by four quantum numbers四个量子数



Section 2 Basic Principles

1 Quantum energy levels & diagrams

2. Atoms with several outer electrons原子外层有多个电子的运动状态

Because the extranuclear electrons interact, the state is described by the principal quantum number主量子数n, the total angular quantum number总角量子数L, the total spin quantum number总自旋量子数S and the total inner quantum number总内量子数J. Transitions normally involve valence electrons发生跃迁的电子一般为价电子.

Term symbol光谱项符号: $n^{2S+1}L_J$

(1) Principal quantum number n: determines the energy of the electron and its mean distance from the nucleus决定电子的能量的高低和核外电子距离核的远近

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

2. Atoms with several outer electrons - optional reading

(2) Total angular quantum number总角量子数L: describes the orbital shape of the outer valence electrons表示外层价电子的轨道形状

It is the vector sum of the angular quantum numbers of the valence electrons其数值为外层价电子角量子数*l*的矢量和: $L = \Sigma l_i$

L = 0, 1, 2, 3, ...

Corresponding symbols: S, P, D, F, ...

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

2. Atoms with several outer electrons - optional reading

(3) Total spin quantum number总自旋量子数S: the vector sum of the spin quantum numbers m_s of the valence electrons外层价电子自旋量子数m_s的矢量和

$S = \Sigma ms,i$ $ms = \pm 1/2, \pm 1, ...$

- For an even number偶数of electrons S = 0 or a positive integer: 0, 1, ...
- For an odd number奇数of electrons S = a positive half-integer: 1/2, 3/2, ...

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

2. Atoms with several outer electrons - optional reading

(4) Inner quantum number内量子数J: the vector sum of L and S外层价电子的总角量子数L和总自旋量子数S的矢量和, taking the values:

$J = |L - S|, |L - S| + 1, |L - S| + 2 ..., L + S$

Number of J values:

- if $L \geq S$, J has (2S+1) values
- if $L < S$, J has (2L+1) values

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

3. Atomic energy-level diagram原子能级图

Energy levels are labelled by term symbols原子能级用光谱项来表征

Example: term symbols from the electron configuration根据原子的电子构型求光谱项

(1) Na ground state:

valence electron价电子of Na: $(3s)^1$

$n = 3$

$L = \sum li = 0$

$S = \sum m_s = 1/2$

$J = 1/2$

Term symbol: $3^2S_{1/2}$

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

(2) First excited state of Na钠原子的第一激发态: $(3p)^1$

$n = 3$

$L = \Sigma l_i = 1$, so L = 0, 1, 2...

$S = 1/2$

$J = |L - S|, |L - S| + 1, |L - S| + 2 ..., (L + S)$

J = 3/2 or 1/2, so (2S+1) = 2

Term: 3^2P fine-structure terms: $3^2P_{1/2}$ and $3^2P_{3/2}$

Because of spin-orbit interaction these two fine-structure terms are two levels of slightly different energy

1 Quantum energy levels & diagrams

Term symbol光谱项符号: $n^{2S+1}L_J$

谱线多重性Multiplicity: 2S+1

Decay from the first excited state to the ground state gives two lines

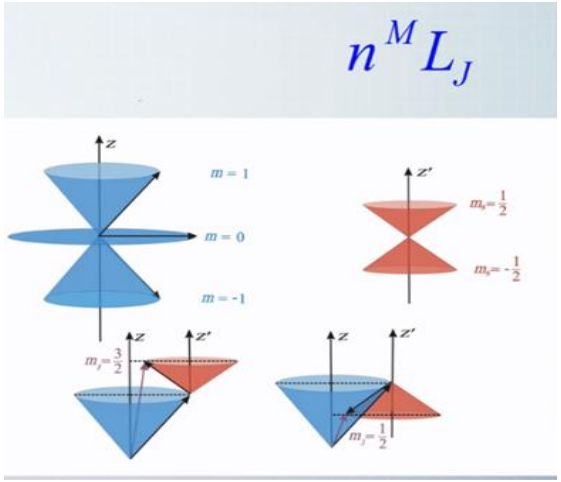
钠原子由第一激发态向基态跃迁发射两条谱线

ground term　　first excited fine-structure terms

$3^2S_{1/2}$ ⟷ $3^2P_{1/2}$

$3^2S_{1/2}$ ⟷ $3^2P_{3/2}$

589.0 nm, 589.6 nm



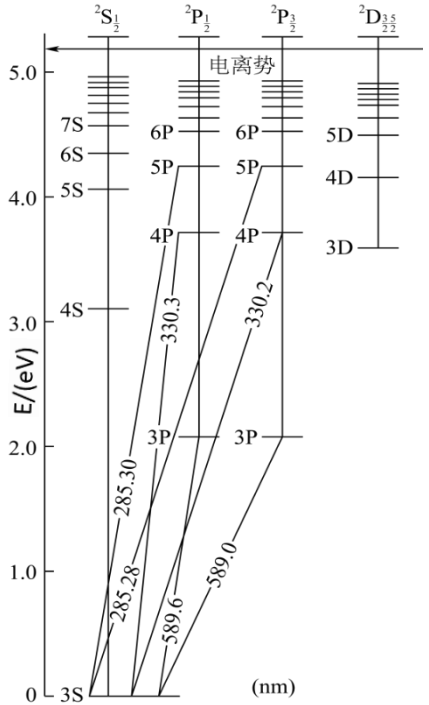
1 Quantum energy levels & diagrams

(3) Energy-level diagram能级图: a graphical display of all possible energy states and transitions把原子中所有可能存在的能级状态及能级跃迁用图解的形式表示.
The vertical axis is the atomic energy E (ground state E = 0), the horizontal axis the fine-structure terms actually present以横坐标表示实际存在的光谱支项. Transitions are drawn as connecting lines series: the set of lines emitted when various upper levels decay to the same lower level各能级之间产生的跃迁用线连接线系：由各种高能级跃迁到同一低能级时发射的一系列光谱线

Spectral terms of the sodium atomic doublet:

Na 589.0nm $3^2S_{1/2} \rightarrow 3^2P_{3/2}$

Na 589.6nm $3^2S_{1/2} \rightarrow 3^2P_{1/2}$



Energy-level diagram of Na

1 Quantum energy levels & diagrams

4. Resonance line共振线

- The radiation absorbed when an atom is excited from the ground state is the resonance absorption line; on returning to the ground state the same wavelength is emitted as the resonance emission line. Together they are called the resonance line of the element.原子由基态跃迁到激发态时，所吸收的一定波长的辐射线称为共振吸收线，再跃迁返回基态时，则发射相同波长的辐射线，称为共振发射线，二者统称为元素共振线
- The resonance line is the most sensitive line of the element. The first resonance absorption line is normally used as the analytical line in quantitative analysis. 共振线是元素所有谱线中最灵敏的谱线。常用元素最灵敏的第一共振吸收线作为分析线。为各元素特征谱线，实际常用定量分析

2 Distribution in energy levels

Quantitative relation between the ground-state atoms and the total number of analyte atoms in the vapour. At thermal equilibrium the populations of the excited state (N_j) and the ground state (N_0) obey the Boltzmann distribution原子化过程中，原子蒸气中基态原子与待测元素原子总数之间的定量关系。热力学平衡时，激发态的原子数Nj和基态的原子数N0的关系符合玻尔兹曼分布定律

Section 2 Basic Principles

2 Distribution in energy levels

1. Boltzmann equation

$$\frac{N_j}{N_0} = \frac{g_j}{g_0} e^{\left(-\frac{E_j - E_0}{KT}\right)}$$

了解：g=2J+1，J 是该能级的总角动量量子数，代表同一个能量能级里，能量相等的微观量子态数目； g_j/g_0 是这条共振跃迁对应的激发态、基态这一对能级的简并度之比。

g_j, g_0: statistical weights of the excited and ground states激发态和基态的统计权重

E_j - E_0: excitation energy激发能

T: absolute temperature绝对温度

K: Boltzmann constant玻尔兹曼常数(1.38×10^{-23} J/K)

2 Distribution in energy levels

Population ratio N_j/N_0 for some elements **(不用背)**

Resonance line /nm	g_j/g_0	ΔE_j /eV	N_j/N_0		
			T=2000K	T=2500K	T=3000K
Na 589.0	2	2.104	9.86×10^{-6}	1.11×10^{-4}	5.83×10^{-4}
Ca 422.7	3	2.932	1.22×10^{-7}	3.67×10^{-6}	3.55×10^{-5}
Mg 285.2	3	4.346	3.35×10^{-11}	5.02×10^{-9}	1.50×10^{-7}
Pb 283.3	3	4.375	2.83×10^{-11}	4.55×10^{-9}	1.34×10^{-7}
Zn 213.9	3	5.795	6.22×10^{-15}	6.22×10^{-12}	5.50×10^{-10}

At < 3000 K, N_j/N_0 for the strongest resonance lines of most elements is only about 10^{-3} (N_0 < 0.1%), so excited atoms can be neglected激发态原子数可以忽略

2 Distribution in energy levels

要避免N_j/N_0太大

2. Factors affecting the distribution原子在各能级分布情况的影响因素

(1) Excitation energy: at the same temperature, the smaller the excitation energy, the larger N_j/N_0激发能：相同温度下，激发能越小，N_j/N_0越大

(2) Temperature: the higher the atomisation temperature, the larger N_j/N_0

温度：原子化温度越高，N_j/N_0越大

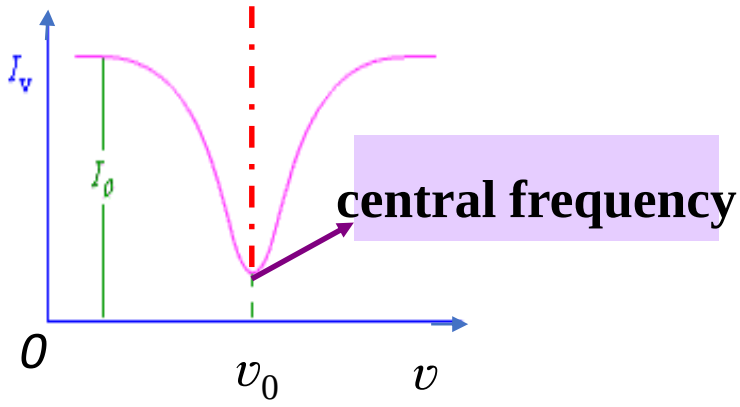
All absorption takes place from the ground state - an important reason for the high sensitivity and the good tolerance of AAS to interferences所有的吸收都是在基态进行的，这是原子吸收分光光度法灵敏度高、抗干扰能力强的一个重要原因

3 Profile and broadening of atomic absorption lines原子吸收线的轮廓和变宽

(1) Profile of an atomic absorption line

原子吸收线的轮廓

Plotting the transmitted intensity I_ν against frequency ν (or wavelength) shows a minimum at the central frequency ν_0, i.e. maximum absorption若将透过光强（I_υ）对频率υ（或波长）作图，由图可见，在中心频率υ_0处透过光强度最小，即吸收最大



Profile of an absorption line

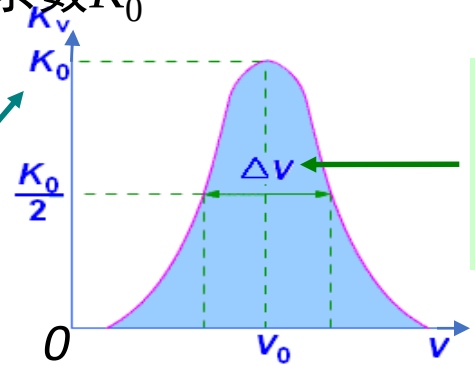
3 Profile and broadening of atomic absorption lines

(1) Profile of an atomic absorption line
原子吸收线的轮廓

Plotting Kν against ν: ν_0 and Δν describe the profile; K_0 is the peak value

将吸收系数K_ν对频率υ作图：υ_0和△υ表征吸收线总体轮廓，其谱线特征包括峰值吸收系数K_0

peak absorption



half width ≈10^{-3} nm

Profile of an absorption line原子吸收线的谱线轮廓

光强I_0与透过光的强度I_ν遵循Lambert定律

$\frac{I_\nu}{I_0} = e^{-K_\nu l}$，l为原子蒸气厚度，$K_\nu$是吸收系数

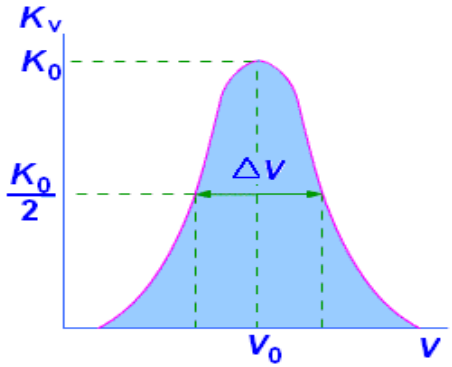
$A = -lg\frac{I_\nu}{I_0} = 0.434K_\nu l$

3 Profile and broadening of atomic absorption lines

(1) Profile of an atomic absorption line

Half width半宽度Δν

- The frequency (or wavelength) difference between the two points of the profile at which the absorption coefficient has fallen to half its peak value ($K_0/2$)指中心频率（或中心波长）的吸收系数一半（$K_0/2$）处所对应的谱线轮廓上两点之间的频率差（或波长差）
- Both ν_0 and $\Delta\nu$ characterise the overall profile
- v_0和$\triangle v$均可表征吸收线的总体轮廓



3 Profile and broadening of atomic absorption lines

(1) Profile of an atomic absorption line

Half width半宽度Δν

The half width depends mainly on:

- the properties of the atom itself, e.g. natural width and isotope effect
- external effects, e.g. thermal, pressure, electric-field, magnetic-field and self-absorption broadening

吸收线的半宽度主要受下列因素的影响：

- 由原子本身的性质决定，如谱线的自然宽度和同位素效应
- 由外界影响引起，如热变宽、压力变宽、电场变宽、磁场变宽、自吸变宽等

3 Profile and broadening of atomic absorption lines

(2) Causes of line broadening谱线变宽的因素

1. Natural width自然宽度

Caused by the nature of the atom. Even without external influences the line still has a finite width - the natural width $\Delta\nu_N$:是由原子本身性质引起。在无外界条件影响时谱线仍有一定的宽度，称为自然宽度（natural width），用△υN表示

In most cases $\Delta\nu_N$ is of the order of 10^{-5} nm and can be neglected多数情况下，$\triangle\upsilon_N$约为10^{-5}nm数量级，可以忽略不计

3 Profile and broadening of atomic absorption lines

(2) Causes of line broadening

2. Doppler broadening多普勒变宽

Doppler broadening arises from the random thermal motion of the atoms, hence it is also called thermal broadening, $\Delta\nu_D$:是由于原子在空间作无规则热运动引起，所以又称为热变宽，用$\triangle v_D$表示

It affects both the emitting atoms in the hollow-cathode lamp and the absorbing ground-state atoms in the atomiser. Its magnitude is of the order of 10^{-3} nm, making it one of the main causes of broadening由于基态原子受热后无规则运动引起的，这种效应无论是在空心阴极灯中发光原子还是原子化器中被测基态原子都存在。热变宽引起的宽度为10^{-3}nm数量级，是谱线变宽的一种主要因素

3 Profile and broadening of atomic absorption lines

(2) Causes of line broadening

2. Doppler broadening

$$\Delta\nu_D = 7.16\times10^{-7}\nu_0\sqrt{\frac{T}{A}}$$

The Doppler width is inversely proportional to the square root of the relative atomic mass, proportional to the square root of the flame temperature, and depends on the line frequency. The smaller the atomic mass and the higher the temperature, the larger $\Delta\nu_D$从上式看出，多普勒变宽与吸收原子自身的相对原子质量的平方根成反比，与火焰的温度平方根成正比，与谱线频率有关被测元素的原子量越小，被测温度越高，$\triangle\upsilon_D$越高

3 Profile and broadening of atomic absorption lines

(2) Causes of line broadening

3. Pressure broadening压力变宽

Pressure broadening: under a given vapour pressure, collisions between the absorbing atoms and other particles (molecules, atoms, ions, electrons) perturb the energy levels and shift the frequency of the emitted or absorbed quanta. It increases with pressure, since collisions become more frequent; its magnitude is about 10^{-3} nm是由于在一定压力蒸气下吸光原子与蒸气中其他粒子（分子、原子、离子和电子）间相互碰撞而引起能级的微小变化，使发射或吸收的光量子频率改变而导致的变宽。这种变宽与吸收区气体的压力有关，压力升高时，粒子间相互碰撞概率增大，谱线变宽严重。其变宽数值约为10^{-3}nm数量级

3 Profile and broadening of atomic absorption lines

(2) Causes of line broadening

3. Pressure broadening

Depending on the collision partner, two types are distinguished:
根据与其碰撞粒子的不同，又可分为下列两种
(1) Holtsmark (resonance) broadening: collisions between excited and ground-state atoms of the analyte; it grows with the atom concentration in the vapour, $\Delta\nu_R$
霍尔兹马克变宽：又称共振变宽，是被测元素激发态原子与基态原子间碰撞引起的谱线变宽，它随试样原子蒸气浓度增加而增加。用$\triangle v_R$表示

3 Profile and broadening of atomic absorption lines

3. Pressure broadening

(2) Lorentz broadening劳伦茨变宽: collisions between analyte atoms and other particles (atoms, molecules, ions, electrons), $\Delta\nu_L$。被测元素原子与其他粒子（原子、分子、离子、电子）相互碰撞而引起的谱线变宽:

$$\triangle v_L = 2N_A\sigma^2 p\sqrt{\frac{2}{\pi RT}(\frac{1}{A}-\frac{1}{M})}$$

N_A为阿福伽德罗常数

σ^2：原子和其他粒子碰撞的有效截面积

P：外界气体压力；A为待测元素的相对原子量

M：为外界气体的相对分子质量；T为热力学温度

R：气体常数（8.31J/mol·K ）

3 Profile and broadening of atomic absorption lines

Under ordinary conditions:

- the profile is dominated by Doppler and Lorentz broadening吸收线轮廓主要受多普勒变宽（温度不能太高）与劳伦茨变宽（压力不能太大）影响；另外相对原子量不能太小
- Broadening generally lowers the sensitivity of the determination谱带变宽往往会导致原子吸收分析的灵敏度降低

4 Absorbance and atom concentration

(1) Integrated absorption积分吸收

Unlike a molecular absorption band, the atomic line profile comes from ground-state atoms of one kind absorbing their resonance radiation, and every point of the profile belongs to the same transition. The concentration of ground-state atoms is therefore proportional to the area under the profile - the integrated absorption与分子吸收光谱不同的是，原子吸收线轮廓是同种基态原子在吸收其共振辐射时被展宽的吸收带，原子吸收线轮廓上的任意各点都与相同的能级跃迁相联系。因此，基态原子浓度与吸收线轮廓所包含的面积（称为积分吸收）呈成正比

Section 2 Basic Principles

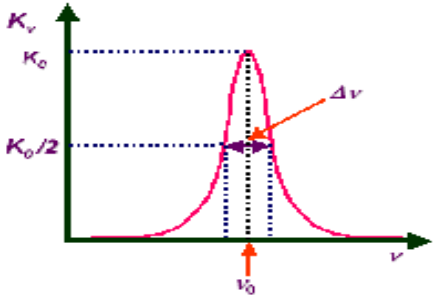
4 Absorbance and atom concentration

(1) Integrated absorption积分吸收

From classical dispersion theory the integrated absorption is:

(不用背)

$$\int_{-\infty}^{+\infty} K_\nu \mathrm{d}\nu = \frac{\pi e^2}{mc} N_0 f \approx K N_0$$



Profile of an absorption line

- If the left-hand side - the area under the line (integrated absorption) - is evaluated,如果将公式左边求出，即谱线下所围面积（积分吸收）
- The number of ground-state atoms N_0 that absorb radiation in the atomic vapor per unit volume can then be obtained.即可得到单位体积原子蒸气中吸收辐射的基态原子数N_0

4 Absorbance and atom concentration

(2) Peak absorption峰值吸收

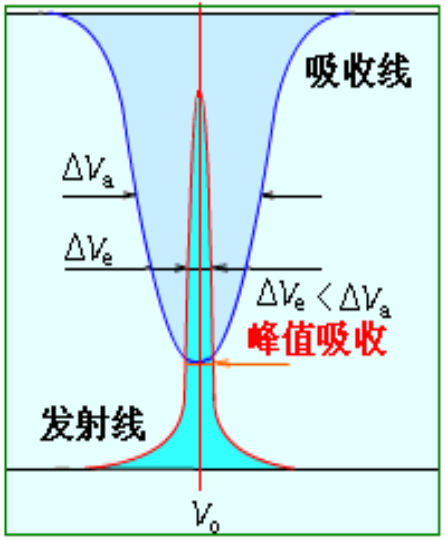
In 1955 A. Walsh proposed using a sharp-line source and measuring the peak absorption instead of the integrated absorption采用锐线光源作为辐射源，用峰值吸收代替积分吸收

Requirements for a sharp-line source:

(1) the emission line coincides with the absorption line (same ν_0)光源的发射线与吸收线的ν_0一致

(2) the half width of the emission line is smaller than that of the absorption line发射线的$\Delta\nu_{1/2}$小于吸收线的$\Delta\nu_{1/2}$

Sharp-line source: hollow-cathode lamp提供锐线光源：空心阴极灯



Measurement of peak absorption

4 Absorbance and atom concentration

With a sharp-line source, K_ν can be treated as constant over the emission-line width and approximately equal to the peak value K_0采用锐线光源进行测量，在辐射线宽度范围内，K_ν可近似认为不变，并近似等于峰值时的吸收系数K_0

$$K_0 = \frac{2}{\Delta\nu}\sqrt{\frac{ln2}{\pi}}\int K_\nu d\nu = \frac{2}{\Delta\nu}\sqrt{\frac{ln2}{\pi}}KN \quad \text{(不用背)}$$

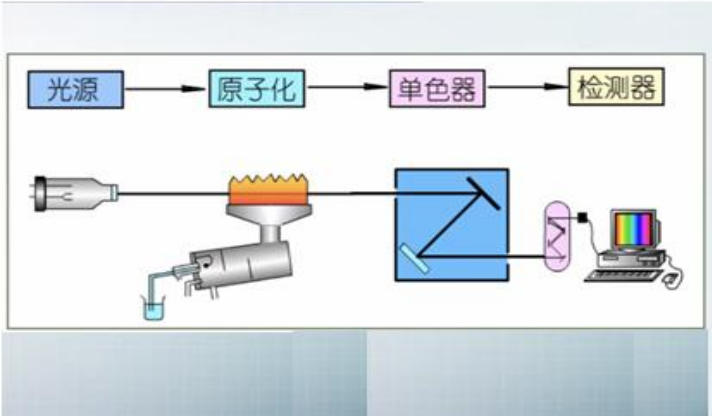
$$A = 0.434K_\nu l = 0.434\ \frac{2}{\Delta\nu}\sqrt{\frac{ln2}{\pi}}KNl \quad \text{(不用背)}$$

$A=K'c$，峰值吸光度与试样中被测元素的浓度呈线性关系，这是原子吸收光谱法的定量分析基础

Section 3 Instrumentation

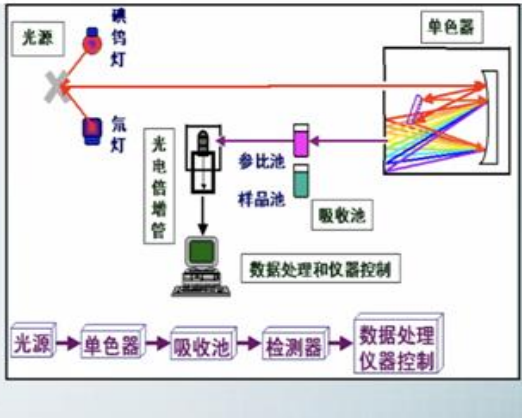
Atomic absorption spectrometer

原子吸收分光光度计



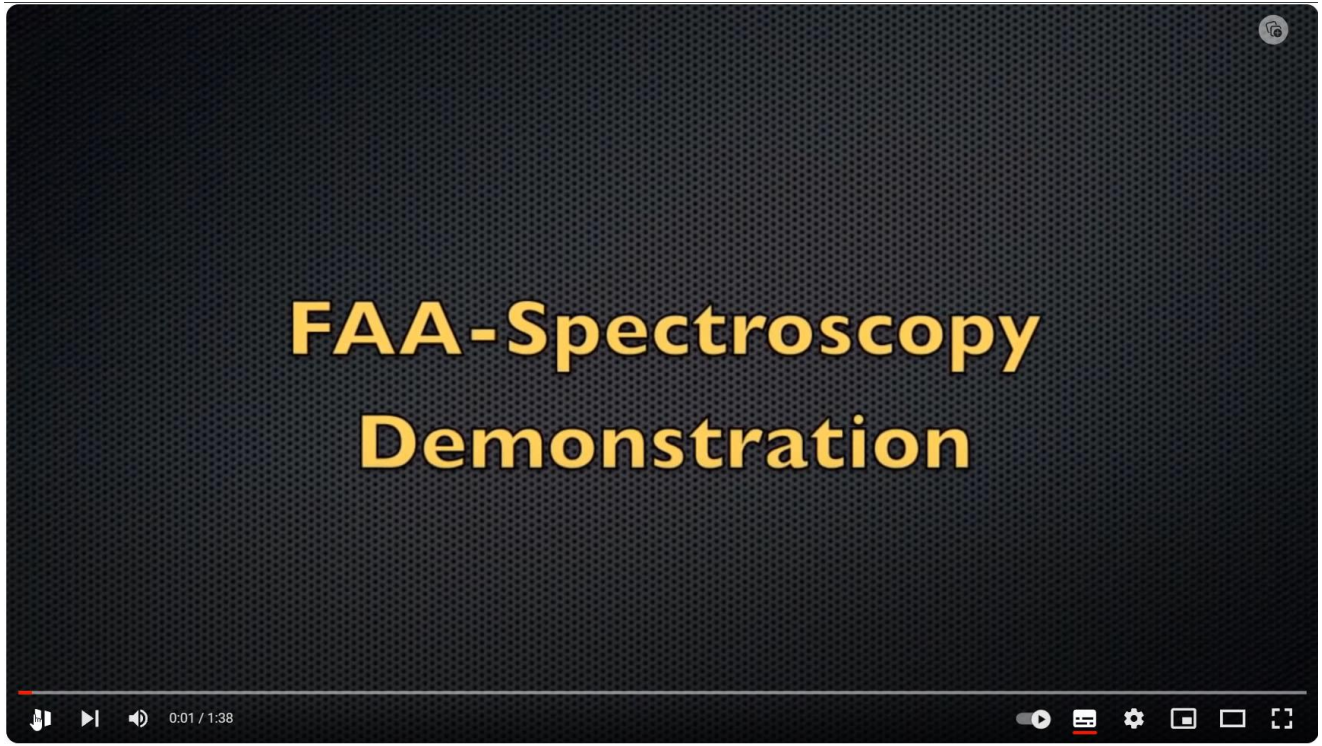
UV-Vis spectrophotometer

紫外-可见分光光度计



- A sharp-line light source is used instead of a continuous light source.用锐线光源代替了连续光源
- An atomizer is used instead of an absorption cell.用原子化器代替了吸收池

Section 3 Instrumentation



Section 3 Instrumentation



1 Components of the instrument

(1) Light source光源

Function: to provide the characteristic spectrum of the analyte, giving high sensitivity and accuracy提供被测元素的特征光谱。获得较高的灵敏度和准确度

Requirements for the source:

(1) emits the resonance line of the analyte能发射被测元素的共振线

(2) emits sharp lines能发射锐线

(3) high radiant intensity and good stability辐射光强度大，稳定性好

Common sources: hollow-cathode lamp, electrodeless discharge lamp, etc.

常用的光源：空心阴极灯，无极放电灯等

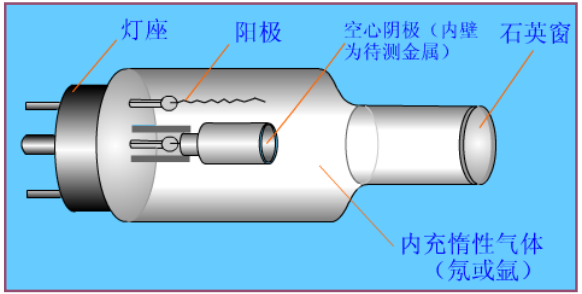
Section 3 Instrumentation

1 Components of the instrument

(1) Light source

1. Hollow-cathode lamp空心阴极灯

(1) Construction: a sealed gas-discharge tube. The cylindrical hollow cathode is made of the pure metal or an alloy of the analyte; the anode is W, Ti or Zr. The tube is filled with Ne or Ar at a few hundred pascals. A quartz window is used below 370.0 nm, optical glass above it.
构造：它是一个封闭的气体放电管。用被测元素纯金属或合金制成圆柱形空心阴极，用钨、钛或锆做成阳极。灯内充Ne或Ar惰性气体，压力为数百帕。发射线波长在370.0nm以下的用石英窗口，370.0nm以上的用光学玻璃窗口



hollow-cathode lamp

Section 3 Instrumentation

1 Components of the instrument



Hollow-cathode lamps

Section 3 Instrumentation

1 Components of the instrument

(1) Light source

1. Hollow-cathode lamp

(2) Principle: when a suitable voltage is applied, electrons travel from the inner wall of the cathode to the anode and ionise the noble gas. The positive ions bombard the cathode and sputter metal atoms from its surface; these atoms are excited by collisions with electrons, gas atoms and ions, and emit the characteristic resonance lines on returning to the ground state.空心阴极灯的原理：施加适当电压时，电子将从空心阴极内壁流向阳极；与充入的惰性气体碰撞而使之电离，产生正电荷，其在电场作用下，向阴极内壁猛烈轰击；使阴极表面的金属原子溅射出来，溅射出来的金属原子再与电子、惰性气体原子及离子发生撞碰而被激发，激发态的原子在返回基态时，发射出相应元素的特征共振线

Section 3 Instrumentation

1 Components of the instrument

(1) Light source

1. Hollow-cathode lamp

$Ar + e \rightarrow Ar^+$

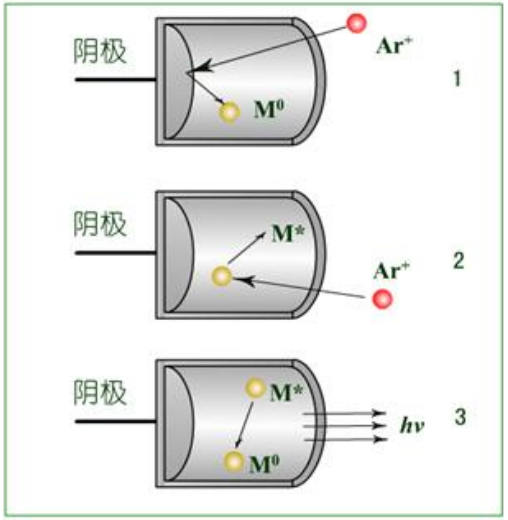
Ar^+ bombards the metal cathode $\rightarrow M + Ar^+$

$M+e \rightarrow M^*+e \quad M^* \rightarrow M + h\nu$

空心阴极灯的原理简述：$Ar + e \rightarrow Ar^+$

Ar^+撞击金属阴极 $\rightarrow M+ Ar^+$

$M+e \rightarrow M^*+e \quad M^* \rightarrow M + h\nu$



Advantages and drawbacks:

(1) high radiant intensity, stable emission, narrow lines

(2) a separate lamp is needed for each element, which is inconvenient

（1）辐射光强度大、稳定、谱线窄

（2）每测一种元素需更换相应的灯，使用不方便

Section 3 Instrumentation

1 Components of the instrument

(1) Light source

2. Multi-element hollow-cathode lamp多元素空心阴极灯

Several metal powders are mixed in suitable proportions, pressed and sintered to form the cathode. Such lamps suffer strong spectral interference and are expensive, so their use is limited.将多种金属粉末按一定的比例混合并压制和烧结，作为阴极，可制得多元素空心阴极灯。 但多元素灯谱线干扰大，价格贵，使用受限制

The lamp current is usually 1-20 mA; choose a suitable value and, when the requirements are met, use the lower current灯工作电流一般在1～20mA，根据情况选择合适的灯电流，满足要求的情况下，选用较低的工作电流

Section 3 Instrumentation

1 Components of the instrument

(2) Atomiser原子化器

Function: to convert the analyte in the sample into gaseous ground-state atoms将试样中被测元素变成气态的基态原子

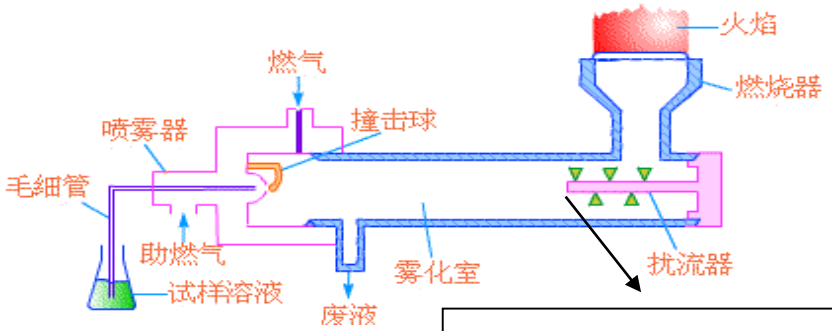
- 四种类型
  - 火焰原子化器(flame atomizer)
  - 石墨炉原子化器(graphite furnace atomizer)
  - 氢化物发生原子化器(hydrogen generation atomizer)
  - 冷蒸汽原子化器(cold atom atomizer)

Section 3 Instrumentation

1 Components of the instrument

(2) Atomiser

1. Flame atomiser火焰原子化器: a chemical flame supplies the energy for atomisation. The premix atomiser is usual; it consists of nebuliser, spray chamber and burner.由化学火焰提供能量，使被测元素原子化。常用的是预混合型原子化器，它包括雾化器、雾化室和燃烧器三部分



premixing and removal of large droplets

Flame atomiser

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

- The nebuliser converts the test solution into a fine, uniform aerosol. The finer the droplets, the more ground-state atoms are formed in the flame.
- 雾化器（nebulizer）的作用是将试液雾化，并使雾滴均匀化。雾滴越小，火焰中生成的基态原子就越多

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

- The spray chamber (1) lets large droplets settle and drain away; (2) mixes the droplets uniformly with fuel and oxidant to form an aerosol for the flame; (3) buffers the gas pressure so that a steady flame is obtained.
- 雾化室（atomizing chamber）的作用：一是使较大雾粒沉降，凝聚从废液口排除；二是使雾粒与燃气、助燃气均匀混合形成气溶胶，再进入火焰原子化；三是起缓冲稳定混合气气压的作用，以便使燃烧气产生稳定的火焰

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

- The burner produces the flame in which the aerosol is evaporated and atomised; a single-slot burner is usually used
- 燃烧器（burner）的作用是产生火焰，使进入火焰的试样气溶胶蒸发、化合等，常用的是单缝燃烧器

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

- Flame: the energy source for atomisation. Dehydration, vaporisation and dissociation of the sample directly determine the atomisation efficiency
- 火焰：进行原子化的能源。试样的脱水、汽化、离解成基态原子直接影响原子化程序

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

- Too high a temperature excites or ionises the atoms, decreasing the ground-state population and the absorbance; too low a temperature fails to dissociate the salts in the sample. The flame temperature must therefore be chosen to suit the element.
- 温度过高，会使试样原子激发或电离，基态原子数减少，吸光度下降。温度过低，不能使试样中盐类解离或解离太小。因此根据情况选择合适的火焰温度

Section 3 Instrumentation

1 Components of the instrument

1. Flame atomisation

Choice of flame temperature火焰温度的选择:

(a) use the lowest temperature that still dissociates the analyte completely into ground-state atoms

(b) the higher the flame temperature, the more thermally excited atoms are produced

(c) the temperature depends on fuel and oxidant; the air-acetylene flame reaches about 2600 K and allows more than 35 elements to be determined

（a）保证待测元素充分离解为基态原子的前提下，尽量采用低温火焰

（b）火焰温度越高，产生的热激发态原子越多

（c）火焰温度取决于燃气与助燃气类型，常用乙炔—空气火焰最高温度2600K左右，能测35以上种元素

Section 3 Instrumentation

1 Components of the instrument

Flames and their temperatures

Flame	Reaction	T_{max} /K
C_3H_8-air	$C_3H_8 + 5O_2 = 3CO_2 + 4H_2O$	2200
H_2-air	$2H_2 + O_2 = 2H_2O$	2300
C_2H_2-air	$2C_2H_2 + 5O_2 = 4CO_2 + 2H_2O$	2600
C_2H_2-N_2O	$C_2H_2 + 5N_2O = 2CO_2 + H_2O + 5N_2$	3200

The flame atomiser is simple to operate, reproducible and fairly sensitive, but its main drawback is a low atomisation efficiency, about 10%

火焰原子化器具有操作简单、重现性好、灵敏度较高等特点，但它的主要缺点是原子化效率低，约为10%

Section 3 Instrumentation

1 Components of the instrument

2. Graphite furnace atomisation石墨炉原子化

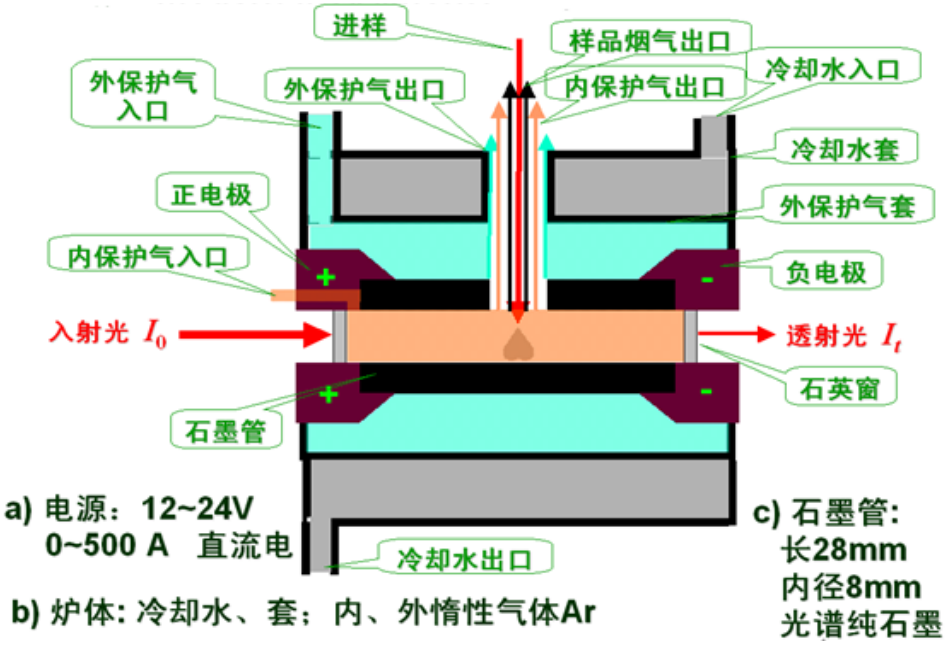
An electrically heated graphite tube containing the sample is raised to a high temperature, atomising the sample是一种电加热器，利用电能加热盛放试样的石墨管，使之达到高温以实现试样原子化

Section 3 Instrumentation

1 Components of the instrume

2. Graphite furnace atomisation石墨炉原子化

(1) Construction: the parts are shown in the figure



Graphite furnace atomiser

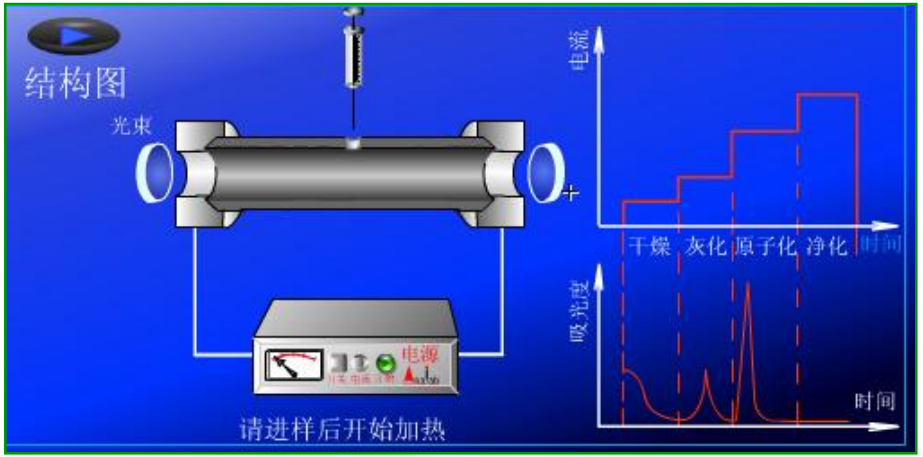
Section 3 Instrumentation

1 Components of the instrument

2. Graphite furnace atomisation

(2) Programme: drying, ashing (removal of the matrix), atomisation and cleaning (removal of the residue); the analyte forms ground-state atoms at high temperature

原子化过程：原子化过程分为干燥、灰化（去除基体）、原子化、净化（去除残渣） 四个阶段，待测元素在高温下生成基态原子



Graphite furnace programme

Section 3 Instrumentation

1 Components of the instrument

2. Graphite furnace atomisation

(2) Programme:

Drying: to evaporate the solvent or the more volatile components of the sample

Ashing: to remove the higher-boiling matrix without loss of the analyte, or to pyrolyse matrices such as fats and oils

Atomisation: a high-power pulse applied to the tube atomises the residue

Cleaning: a higher temperature removes residual matter from the tube

干燥：目的是蒸发除去溶剂，或样品中挥发性较大的组分

灰化：目的是在不损失被测元素的前提下，将沸点较高的基体蒸发除去，或是对脂肪和油等基体物质进行热解

原子化：施加大功率于石墨炉上，使待测残渣受到突然的功率脉冲，从而原子化

净化：用较高温度除去残留在管内的残渣

Section 3 Instrumentation

1 Components of the instrument

方法	火焰原子化法	石墨炉原子化法
原子化热源	化学火焰能	电热能
原子化温度	相对较低	相对较高
原子化效率	较低（<30%）	较高（>90%）
进样体积	较多（1~5 mL）	较少（1~50 μL）
检出限	高Cd:0.5 ng/mL Al:20 ng/mL	低Cd:0.002 ng/mL Al:1.0 ng/mL
重现性	较好，RSD为0.5%~1%	较差，RSD为1.5%~5%
基体效应	较小	较大

从成本考虑，火焰法；对灵敏度有要求，石墨炉

Section 3 Instrumentation

1 Components of the instrument

3. Hydride generation atomiser氢化物发生原子化器

It consists of a hydride generator and an atomisation device and is a low-temperature technique. For elements such as Hg, Ge, Sn, Pb, As, Sb, Bi, Se and Te, liquid sampling with either a flame or a graphite furnace gives poor sensitivity.

由氢化物发生器和原子化装置二部分组成，属于低温原子化技术。有一些元素（如Hg、Ge、Sn、Pb、As、Sb、Bi、Se和Te等）采取液体进样时，无论是火焰原子化或石墨炉原子化均不能得到较好的灵敏度

Section 3 Instrumentation

1 Components of the instrument

4. Cold vapor generator冷蒸气发生原子化器

Consists of a cold vapor generator and an atomic absorption cell由冷蒸气发生器和原子吸收池组成

Designed specifically for mercury determination专门用于汞的测定

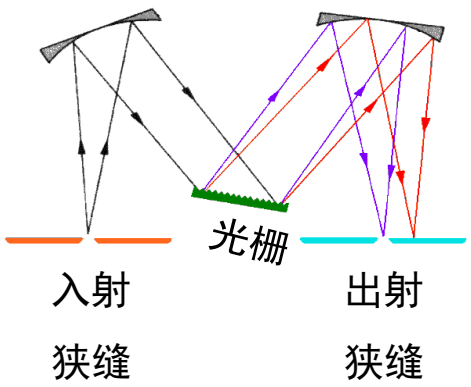
Inorganic mercury: reduced to Hg by $SnCl_2$ in acidic solution无机汞：酸性溶液$SnCl_2$还原为Hg

Organic mercury: reduced by $SnCl_2$ after pretreatment and atomized at ambient temperature and pressure有机汞：预处理后，$SnCl_2$还原，常温常压原子化

Carrier gas (Ar, N_2, air) transports mercury atomic vapor into the absorption cell载气（Ar，N_2，空气）将汞原子蒸气送入吸收池

Section 3 Instrumentation





1 Components of the instrument

(3) Optical system

- Function: to separate the required resonance absorption line from neighbouring interfering lines作用：将所需的共振吸收线与邻近干扰线分离
- To prevent other radiation from the atomiser from reaching the detector indiscriminately and to avoid fatigue of the photomultiplier, the monochromator is placed after the atomiser为了防止原子化时产生的其他辐射不加选择地都进入检测器，以及避免光电倍增管的疲劳，单色器通常配置在原子化器之后
- The key component is the dispersing element, nowadays a grating
- 单色器中的关键部件是色散元件，现多用光栅

Section 3 Instrumentation

1 Components of the instrument

(4) Detection system

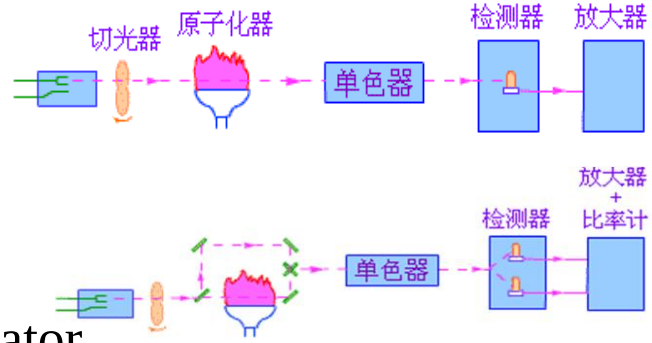
- It consists of a detector, an amplifier and a data-processing system主要由检测器、放大器、数据处理系统组成
- The detector converts the optical signal from the monochromator into an electrical signal; a photomultiplier tube is usual检测器的作用是将单色器分出的光信号转换成电信号，常用光电倍增管

Section 3 Instrumentation

2 Types of instrument

1. Single-beam spectrometer

2. Double-beam spectrometer



A chopper splits the resonance beam into two: the measuring beam S passes through the atomiser and the reference beam R does not. They enter the monochromator alternately and the difference in output is measured. The reference beam cancels drift of the source and detector but not instability of the atomiser.由光源发射的共振线被斩光器分解成两束光，一束测量光S通过原子化器，另一束光R作为参比不通过原子化器，两束光交替进入单色器；检测系统输出的信号差；参比光束可消除光源和检测器不稳定的影响，但无法消除原子化系统不稳定的误差

Section 3 Instrumentation

2 Types of instrument

Summary

Comparison: UV-Vis absorption versus atomic absorption

1. Light source

continuum source versus sharp-line source

2. Temperature

room temperature versus high temperature

3. Sample

clear solution versus liquid or solid

4. Spectrum

broad-band absorption versus narrow-band absorption

An atomic absorption line is not a geometrical line but has a finite width, i.e. a range of frequencies

Section 3 Instrumentation

2 Types of instrument

小结

对比： UV-vis吸收光谱 ←→ 原子吸收光谱

1.光源

连续光源 ←→ 锐线光源

2.检测温度

室温 ←→ 高温

3.检测样品

澄清溶液 ←→ 液体或固体

4.谱图

宽带吸收光谱 ←→ 窄带吸收光谱

原子吸收光谱，非几何意义上的线，而是具有一定宽度或频率范围的谱线

Section 4 Quantitation

Section 4 Quantitation

1 Measurement conditions

1. Sample size and pretreatment

Decided by the nature and content of the element, the method used and the required precision根据被测元素性质、含量、分析方法及要求的精密度来确定

2. Choice of analytical line

The main resonance line is normally used because it is the most sensitive absorption line. Lines lying in the far ultraviolet are better avoided.通常选择主共振线作为分析线，因主共振线是最灵敏的吸收线。共振线靠近远紫外的不宜选用

Section 4 Quantitation

1 Measurement conditions

3. Slit width

There are far fewer absorption lines than emission lines, so overlap is unlikely and a wider slit may be used, increasing sensitivity and improving the signal-to-noise ratio吸收线数目比发射线数目少的多，谱线重叠干扰的概率小，因此，允许使用较宽的狭缝，有利于增加灵敏度，提高信噪比

4. Lamp current

Use the lowest current that keeps the discharge stable with sufficient intensity (about one half to two thirds of the rated value)在保证放电稳定和足够光强的条件下，尽量选用低的工作电流（1/2~2/3的工作电流）

Section 4 Quantitation

1 Measurement conditions

5. Atomisation conditions

In flame AAS the choice and adjustment of the flame govern the atomisation efficiency: (1) choose a flame type that does not interfere with the absorption line; (2) easily ionised elements need a cool flame, refractory compounds a hot flame; (3) in graphite furnace AAS the programme (drying, ashing, atomisation, cleaning) must be optimised experimentally在火焰原子化法中，火焰的选择和调节是保证原子化效率的关键：①选择不干扰吸收线的火焰类型；②易电离元素：低温火焰；难离解物质：高温火焰；③石墨炉原子化法中，原子化程序要经过干燥、灰化、原子化和净化几个阶段，各阶段的温度与持续时间均要通过实验选择

Section 4 Quantitation

2 Interferences & Removal

Interferences in AAS: ionisation, matrix, chemical and optical interference

1. Ionisation interference电离干扰

- Phenomenon: the analyte is ionised during atomisation, so fewer ground-state atoms are available for absorption and A decreases现象：被测元素在原子化过程中发生电离，使参与吸收的基态原子数减少而造成吸光度A下降
- Removal: add an ionisation suppressor, i.e. an easily ionised element such as an alkali metal salt
- 消除方法：加入消电离剂（加入易电离元素，如碱金属盐）

Section 4 Quantitation

2 Interferences & Removal

2. Matrix (physical) interference基体干扰（即物理干扰）

- Phenomenon: changes in the physical properties of the sample during handling, transfer, evaporation and atomisation lower the absorbance; a non-selective interference现象：试样在处理、转移、蒸发和原子化过程中，由于试样物理特性的变化引起吸光度下降，是非选择性干扰
- Removal: make the composition of sample and standard solutions as similar as possible消除方法：可通过控制试液与标准溶液的组成尽量一致的方法来消除

Section 4 Quantitation

2 Interferences & Removal

3. Chemical interference化学干扰

- Phenomenon: reactions between the analyte and coexisting components in solution or in the gas phase affect the dissociation of the analyte compound and the atomisation efficiency现象：在溶液或气相中，由于被测元素与其他共存组分之间发生化学反应，而影响被测元素化合物的离解和原子化效率
- Removal: add a releasing agent or a protective agent, or raise the flame temperature消除方法：可加入释放剂、保护剂、适当提高火焰温度等

2 Interferences & Removal

4. Optical interference光学干扰

It includes spectral-line interference and non-absorption-line interference包括光谱线干扰和非吸收线干扰

- Phenomenon (spectral-line interference): an absorption line of a coexisting element lies close to the analytical line, i.e. the lines overlap现象：光谱线干扰是试样中共存元素的吸收线与被测元素的分析线相近（吸收线重叠）而产生的干扰

  e.g. Fe 271.903 nm Pt 271.904 nm

- Removal: choose another sensitive line that is less interfered with, or separate the interferent chemically消除方法：可另选灵敏度高而干扰少的分析线或用化学方法分离干扰元素

Section 4 Quantitation

2 Interferences & Removal

Non-absorption-line interference is a form of background absorption非吸收线干扰是一种背景吸收（background absorption）

- Phenomenon: absorption of the resonance line by molecules, oxides and salts formed in the atomiser, plus scattering of light by fine solid particles现象：原子化过程中生成的气体分子、氧化物、盐类等对共振线的吸收及微小固体颗粒使光产生散射而引起的干扰
- Removal: adjacent-line correction, continuum-source correction (a deuterium lamp in the UV) or Zeeman-effect correction消除方法：邻近线法、连续光源（在紫外光区通常用氘灯）法、塞曼（Zeeman）效应法等

Section 4 Quantitation

3 Sensitivity & detection limit

Characteristic sensitivity特征灵敏度

- In AAS the 1% absorption sensitivity is preferred; it is also called the characteristic sensitivity and is defined as the concentration or mass of the analyte that produces 1% absorption (A = 0.0044)在原子吸收分光光度中，更习惯用1%吸收灵敏度表示，也称特征灵敏度（characteristic sensitivity）。其定义为能产生1%吸收（或吸光度为0.0044）信号时，所对应的被测元素的浓度或被测元素的质量

characteristic concentration

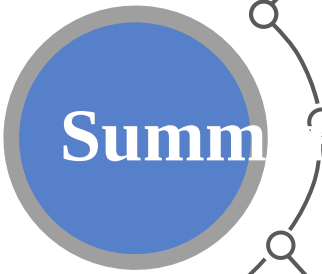
$$s_c = \frac{0.0044 \times c_x}{A}$$

characteristic mass

$$s_m = \frac{0.0044 \times m_x}{A} = \frac{0.0044 \times c_x V}{A}$$

- The smaller the 1% absorption sensitivity, the higher the sensitivity of the method

Chapter summary



1. Overview

atomic and molecular spectrometry, AAS, AES, ICP-AES

2. Basic principles of AAS

Term symbols, energy levels, resonance lines, half width, Doppler and pressure broadening, integrated and peak absorption

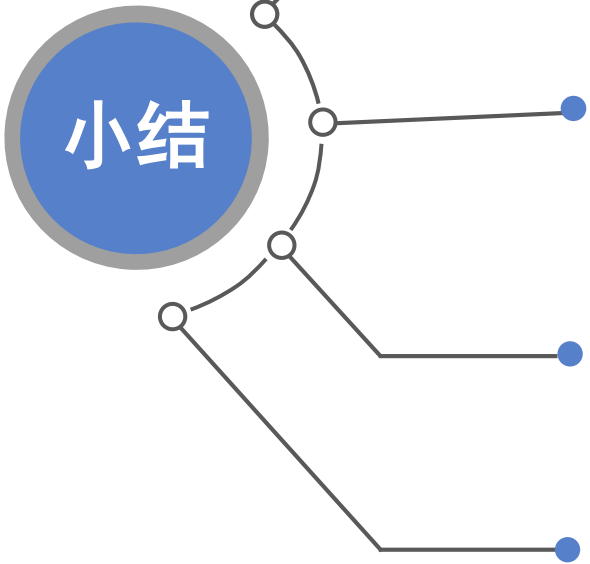
3. Instrumentation

Main parts: light source, atomiser, optical system, detection system

4. Quantitative methods

choice of conditions, interferences and their removal, sensitivity and limit of detection

本章小结



1.概述

原子光谱法和分子光谱法、原子吸收分光光度法、原子发射光谱法、电感耦合等离子体原子发射光谱法

2.原子吸收分光光度法的基本原理

光谱项、原子能级图、共振吸收线、半宽度、多普勒变宽（热变宽）、压力变宽、积分吸收、峰值吸收、原子吸收值与原子浓度的关系$A=K'c$

3.原子吸收分光光度计

原子吸收分光光度计主要组成：光源、原子化器、分光系统、检测系统

4.定量分析方法

测定条件的选择、干扰及消除方法、灵敏度和检出限

定量方法：标准曲线法、标准加入法、内标法

Thank you